VLSI Decimation Filter Design REPORT

1. Objective and Architecture

Convert a 4.096 MHz one-bit sigma-delta stream to 2 kHz. Task 1 uses a CIC decimator; Task 2 adds a short FIR filter to correct CIC droop near 1 kHz.

Signal path

Input (1-bit stream, 4.096 MHz) → 3 integrators → decimation by 2048 → 3 combs → 6-tap FIR compensation (Task 2) → 20-bit signed output at 2 kHz.

2. CIC Settings

Setting	Value	Reason
Order, (N)	3	Best 990 Hz result among tested orders 2, 3 and 4
Decimation, (R)	2048	4.096MHz/2048=2kHz
Differential delay, (M)	1	(M=2) creates a null at 1 kHz, near the 990 Hz tone
Internal width	35 bits	Growth bound: $3\log_2(2048)+2=35$bits

The result is saturated to signed 20-bit output. The design notes report no clipping, a peak near 97% of full scale, and approximately 11.5 dB attenuation at 990 Hz.

3. Six-Tap FIR Compensator

The FIR runs at 2 kHz on full-precision CIC samples. Its coefficients approximate the inverse of CIC passband droop. The design uses a minimum-phase response, truncates it to six taps, and then refines the coefficients using least squares

Selected Coefficients:

Tap	Coefficient
h[0]	182004
h[1]	−96584
h[2]	50848
h[3]	−28297
h[4]	15908
h[5]	−7553

Six taps limit startup-settling effects in the 2000-sample SNR run. The project notes report about 30 dB lower SNR for a 23-tap alternative. The coefficients include output gain, and rounding and saturation occur once, at the end.

4. Reported SNR

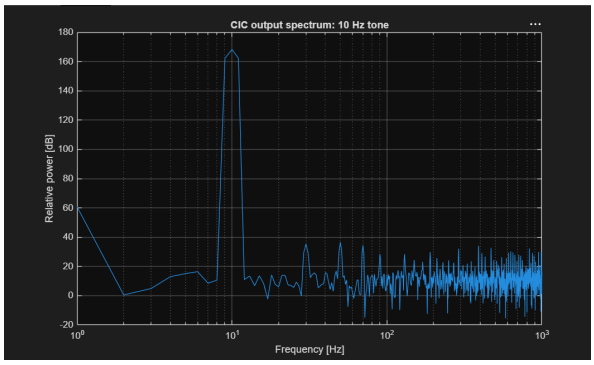
Input tone	CIC only	CIC + FIR	Change
10 Hz	121.00 dB	121.75 dB	+0.75 dB
500 Hz	159.22 dB	138.82 dB	−20.40 dB
990 Hz	111.08 dB	120.34 dB	+9.26 dB

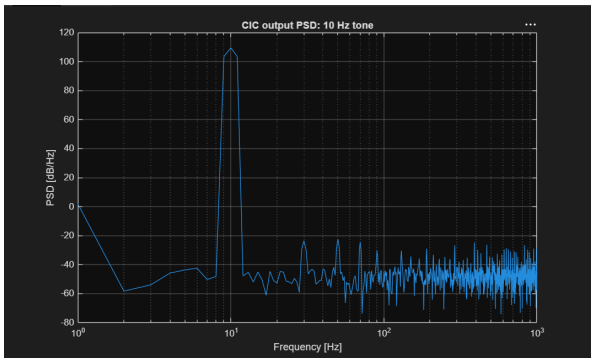
Main result**:** At 990 Hz, SNR improves by 9.26 dB. At 500 Hz, the reported SNR drops by 20.40 dB. The project notes flag this tone as atypical because its four-sample output pattern affects rounding-error cancellation.

5. MATLAB Plots

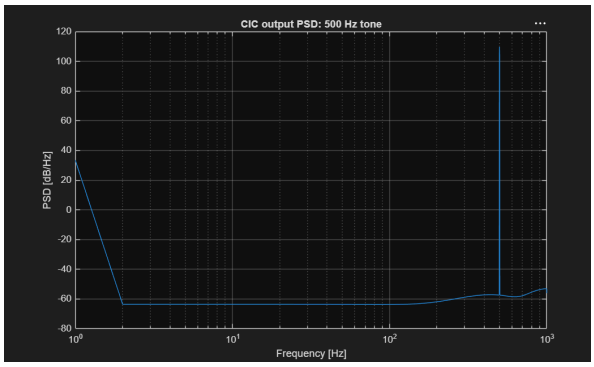
PSD SNR

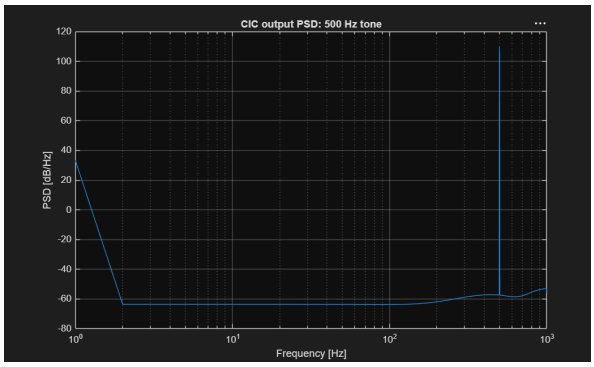
10Hz:



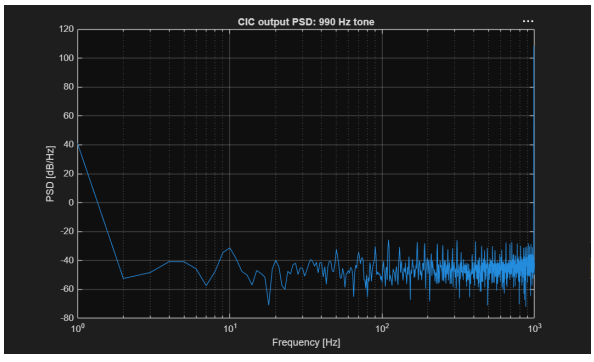


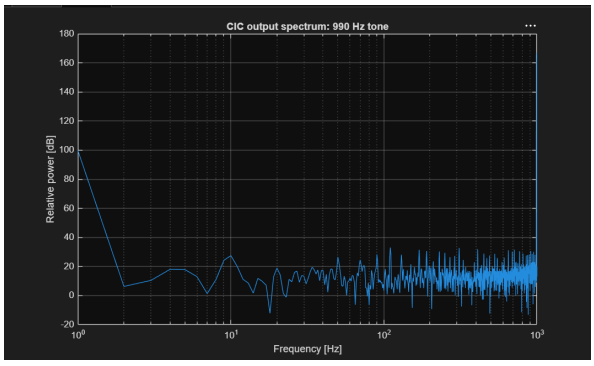
500Hz:





990Hz:





Github Link:

https://github.com/kushalkatekar9-dot/Decimation-Filter-Design